

Random Signal Theory–EEE 554

Lecture Notes 1

Review on Basic Probability

[Reading: Kay, Sections 3.3-3.9, 4.3-4.5]

Concept of probability

- Measure of uncertainty assigned to events or outcomes of random experiments
- Number describing chance or likelihood of event occurring

Methods to obtain probability of an event

Counting method:

assume *finite* number of equally likely outcomes; the probability of an event is obtained by *counting the number of times the event occurs* in proportion to the total number of possible outcomes

Relative frequency method:

assume that an experiment can be repeated an infinite number of times under the same conditions; the probability of an event is obtained by the total number of occurrences of the event relative to the total number of experiments (or trials)

Use of probabilistic models:

describe uncertainty using a *pattern* that the occurrence of the event follows; one such simple model is based on *set theory*

Using the concept of sets as a *probabilistic model*

- ▶ Universal set $\mathcal{S}$: sample space (or certain event)
the elements of $\mathcal{S}$ are **outcomes** or sample points
- ▶ Null set $\emptyset$: impossible event
- ▶ Subset $\mathcal{A} \subset \mathcal{S}$: event whose elements are outcomes from sample space
- ▶ Complement set $\mathcal{A}^c$: all the elements not in event $\mathcal{A}$
- ▶ **Union** set $\mathcal{A} \cup \mathcal{B}$: elements in event **$\mathcal{A}$ or $\mathcal{B}$ or both** events ($\mathcal{A} + \mathcal{B}$)
- ▶ **Intersection** set $\mathcal{A} \cap \mathcal{B}$: elements in event **$\mathcal{A}$ and $\mathcal{B}$** ($\mathcal{A}\mathcal{B}$)
- ▶ Disjoint sets $\mathcal{A}, \mathcal{B}$: mutually exclusive events, no common elements, $\mathcal{A} \cap \mathcal{B} = \emptyset$

Assign probabilities to outcomes or events resulting from a random experiment

- ▶ Random experiment results in outcomes
- ▶ Event is a specific outcome or set of outcomes from the experiment

Example:

experiment is toss a (fair) coin multiple times

outcomes are heads or tails

one possible **event** is $\mathcal{A} = \{\text{heads, heads, tails}\}$ (three specific outcomes when coin is tossed three times)

Example:

experiment is collect data samples

outcomes are N samples with values $x_n, n = 0, 1, \dots, N - 1$

one possible **event** is $\mathcal{A} = \{x_0, x_1, x_2, x_3\}$ (first four samples of the data)

Axioms of Probability

1. $\Pr(\mathcal{A}) \geq 0$, for an event $\mathcal{A} \subset \mathcal{S}$
2. $\Pr(\mathcal{S}) = 1$ (probability of certain event is one)
3. $\Pr(\mathcal{A} \cup \mathcal{B}) = \Pr(\mathcal{A} + \mathcal{B}) = \Pr(\mathcal{A}) + \Pr(\mathcal{B})$ if $\mathcal{A} \cap \mathcal{B} = \mathcal{A}\mathcal{B} = \emptyset$
(i.e., if $\mathcal{A}$ and $\mathcal{B}$ are mutually exclusive)

Properties (that follow from the three axioms of probability)

1. $\Pr(\emptyset) = 0$

To show this, since $\mathcal{S}$ and $\emptyset$ are mutually exclusive, $\Pr(\mathcal{S} + \emptyset) = \Pr(\mathcal{S}) + \Pr(\emptyset)$ (Axiom 3). Also, as the union of $\mathcal{S}$ and $\emptyset$ is $\mathcal{S}$, $\Pr(\mathcal{S} + \emptyset) = \Pr(\mathcal{S})$ thus, $\Pr(\mathcal{S} + \emptyset) = \Pr(\mathcal{S}) + \Pr(\emptyset)$ and $\Pr(\mathcal{S} + \emptyset) = \Pr(\mathcal{S})$ since from Axiom 2, $\Pr(\mathcal{S}) = 1$, then $\Pr(\emptyset)$ must equal 0

2. $\Pr(\mathcal{A}) = 1 - \Pr(\mathcal{A}^c) \leq 1$

3. Two events $\mathcal{A}$ and $\mathcal{B}$ are **equal with probability one** if and only if (iff) they have the same elements; then we can write $\Pr(\mathcal{A}) = \Pr(\mathcal{B}) = \Pr(\mathcal{A} \mathcal{B})$ (intersection)

4. $\Pr(\mathcal{A} + \mathcal{B}) = \Pr(\mathcal{A}) + \Pr(\mathcal{B}) - \Pr(\mathcal{A} \mathcal{B})$

6. **Equally probable events**: If experiment had N possible outcomes, then the outcomes are equally probable if the probability of each outcome is $p_i = \frac{1}{N}$, $i = 1, 2, \dots, N$

7. Bayes' theorem (conditional probability): **[Kay, Section 4.5]**

$$\Pr(\mathcal{A} | \mathcal{B}) = \frac{\Pr(\mathcal{A} \mathcal{B})}{\Pr(\mathcal{B})}, \text{ probability of } \mathcal{A} \text{ given that } \mathcal{B} \text{ has occurred, } \Pr(\mathcal{B}) \neq 0,$$

Variations of this property, based on what information is provided

$$\circ \Pr(\mathcal{A} \mathcal{B}) = \Pr(\mathcal{A} | \mathcal{B}) \Pr(\mathcal{B}) = \Pr(\mathcal{B} | \mathcal{A}) \Pr(\mathcal{A})$$

$$\circ \Pr(\mathcal{A} | \mathcal{B}) = \frac{\Pr(\mathcal{B} | \mathcal{A}) \Pr(\mathcal{A})}{\Pr(\mathcal{B})}$$

○ if $\mathcal{B} \subset \mathcal{A}$, then $\mathcal{A}\mathcal{B} = \mathcal{B}$, and $\Pr(\mathcal{A}|\mathcal{B}) = \frac{\Pr(\mathcal{A}\mathcal{B})}{\Pr(\mathcal{B})} = 1$

○ if $\mathcal{A} \subset \mathcal{B}$

then $\mathcal{A}\mathcal{B} = \mathcal{A}$, and $\Pr(\mathcal{A}|\mathcal{B}) = \frac{\Pr(\mathcal{A}\mathcal{B})}{\Pr(\mathcal{B})} = \frac{\Pr(\mathcal{A})}{\Pr(\mathcal{B})}$

Also, $\Pr(\mathcal{A}|\mathcal{B}) = \frac{\Pr(\mathcal{A})}{\Pr(\mathcal{B})} \geq \Pr(\mathcal{A})$

since $\Pr(\mathcal{B}) \geq \Pr(\mathcal{A})$ and $\Pr(\mathcal{B}) \leq 1$ and $\Pr(\mathcal{A}) \leq 1$

○ $\Pr((\mathcal{A} + \mathcal{B})|\mathcal{D}) = \Pr(\mathcal{A}|\mathcal{D}) + \Pr(\mathcal{B}|\mathcal{D})$ if and only if $\Pr(\mathcal{A}\mathcal{B}) = 0$

follows from distributive property from set theory $\mathcal{D} \cap (\mathcal{A} \cup \mathcal{B}) = (\mathcal{D} \cap \mathcal{A}) \cup (\mathcal{D} \cap \mathcal{B})$

Example of Bayes' theorem

Let $\mathcal{A}$ = you are near the ocean and $\mathcal{B}$ = you picked up a seashell

Since the oceans cover 71% of the Earth's surface, we can assume that $\Pr(\mathcal{A}) \approx 0.6$
also assume that the probability of you picking up a seashell is $\Pr(\mathcal{B}) \approx 0.5$.

It is highly probable that when you are at the ocean, you will pick up a seashell
so, $\Pr(\mathcal{B}|\mathcal{A})$ is pretty high (probability of picking up a seashell given that you are at the ocean)

Using Bayes' rule, we can write this probability as $\Pr(\mathcal{B}|\mathcal{A}) = \frac{\Pr(\mathcal{A}|\mathcal{B}) \Pr(\mathcal{B})}{\Pr(\mathcal{A})}$

where $\Pr(\mathcal{A}|\mathcal{B})$ is probability that you are near the ocean given that you picked up a seashell

Also using Bayes' rule, this probability can be written as: $\Pr(\mathcal{A}|\mathcal{B}) = \frac{\Pr(\mathcal{B}|\mathcal{A}) \Pr(\mathcal{A})}{\Pr(\mathcal{B})}$

From right-hand side of equation, since $\Pr(\mathcal{A}) \approx 0.6$, $\Pr(\mathcal{B}) \approx 0.5$, $\frac{\Pr(\mathcal{B})}{\Pr(\mathcal{A})} = 0.833$ (pretty high),
and since $\Pr(\mathcal{B}|\mathcal{A})$ is large, then $\Pr(\mathcal{A}|\mathcal{B})$ is also large

According to Bayes' rule, if you pick up a seashell (anywhere), you are probably near the ocean; which means, you do not have to follow the the *common belief* that you have to hold the seashell to your ear to hear (and know that you are near) the ocean

Example of Bayes' theorem

Dynamic state space representation with unknown state $\mathcal{X}_k$ at time step k
(with transition modelling error $\mathcal{U}_k$) and observation $\mathcal{Z}_k$
(with additive white Gaussian noise $\mathcal{W}_k$)

$$\mathcal{X}_k = f(\mathcal{X}_{k-1}) + \mathcal{U}_{k-1}$$

$$\mathcal{Z}_k = h(\mathcal{X}_k) + \mathcal{W}_k$$

We need to obtain $\Pr(\mathcal{X}_k | \mathcal{Z}_k)$ using transition equation (predict) and measurement equation (update)

Using Bayes' theorem

$$\Pr(\mathcal{X}_k | \mathcal{Z}_k) = \frac{\Pr(\mathcal{Z}_k | \mathcal{X}_k) \Pr(\mathcal{X}_k)}{\Pr(\mathcal{Z}_k)}.$$

Recursive implementation known as Kalman filter

only applicable for linear systems with Gaussian noise and modelling error